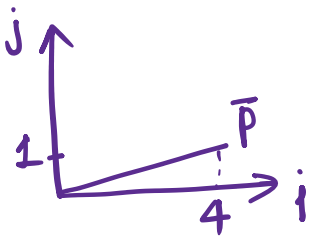
$x[n]$: N-periodic finite power discrete time signal

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n} \longrightarrow \text{F.S. Represent}^n$$



$\Rightarrow \bar{p} = 4i + 1j$

$\bar{p} = \alpha_1 i + \alpha_2 j \quad \alpha_1 = \bar{p}\cdot i \;\&\; \alpha_2 = \bar{p}\cdot j$

$\{x_k\}_{k=0}^{N-1}$ is o.n. set, $x_k[n] = e^{jk\frac{2\pi}{N}n}$

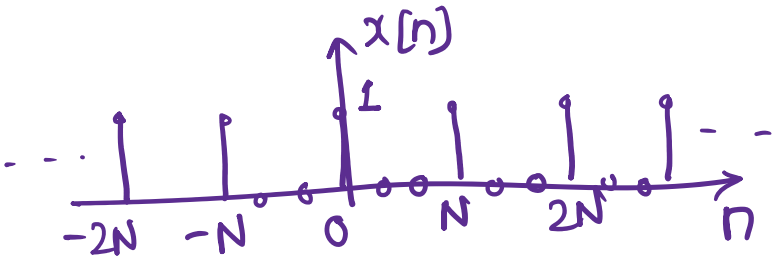
$$\langle x, x_p\rangle = \langle \sum a_k x_k, x_p\rangle = \sum a_k \langle x_k, x_p\rangle = a_p$$

$$\therefore a_k = \langle x, x_k\rangle$$

$$= \frac{1}{N}\sum_{\langle N\rangle} x[n]\, x_k^*[n]$$

$$= \frac{1}{N}\sum_{\langle N\rangle} x[n]\, e^{-jk\frac{2\pi}{N}n}$$

$$\left.\begin{aligned} x[n] &= \sum_{k=\langle N\rangle} a_k e^{jk\frac{2\pi}{N}n} \longrightarrow \text{Synthesis} \\ a_k &= \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-jk\frac{2\pi}{N}n} \longrightarrow \text{Analysis} \end{aligned}\right\} \text{D.T.F.S.}$$



$\Rightarrow$ Period $= N$

$$a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-jk\frac{2\pi}{N}n}$$

$$= \frac{1}{N} \quad \forall k$$

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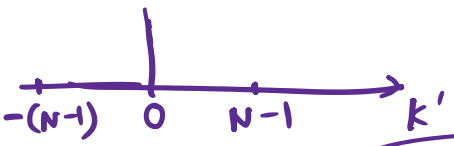
① For real $x[n]$: $a_{-k} = a_k^*$

Pf: $x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$



$x[n]$

Pf: $x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$ $\quad -(N-1) \quad 0 \quad N-1 \quad k' \quad x[n]$

$$x^*[n] = \sum_{k=0}^{N-1} a_k^* e^{-jk\frac{2\pi}{N}n} = \sum_{k'=-(N-1)}^{0} a_{-k'}^* e^{jk'\frac{2\pi}{N}n} = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$$

Let $k' = -k$

$$= \sum_{k=0}^{N-1} a_{-k}^* e^{jk\frac{2\pi}{N}n}$$

$\Rightarrow a_k = a_{-k}^*$

$\Rightarrow a_{-k} = a_k^*$

(II) $a_{k+N} = a_k$

Pf: $a_{k+N} = \sum x[n] e^{-j(k+N)\frac{2\pi}{N}n} = \sum x[n] e^{-jk\frac{2\pi}{N}n} \cdot 1$

$= a_k$

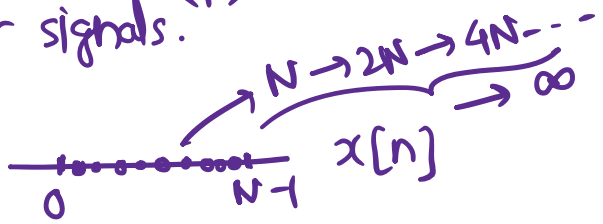
$\Rightarrow a_k$'s are periodic with N.

Continuous Time Fourier Series (CTFS):

$$S_T = \{x : \mathbb{R} \to \mathbb{C} \mid x(t+T) = x(t) \ \forall t, \ \frac{1}{T}\int_{\langle T \rangle} |x(t)|^2 < \infty\}$$

Set of all T-periodic finite power signals.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$



To show: S_T is a V.S. with basis $\{e^{jk\frac{2\pi}{T}t}\}_{k=-\infty}^{\infty}$

(1) S_T is V.S.

x & $y \in S_T$

$(x+y)(t) \triangleq x(t) + y(t)$ & $(\alpha x)(t) \triangleq \alpha x(t)$

$(x+y)(t+T) = x(t+T) + y(t+T)$

$= x(t) + y(t) = (x+y)(t)$

$$\frac{1}{T}\int_{\langle T \rangle} |(x+y)(t)|^2 dt \le \frac{1}{T}\int_{\langle T \rangle} |x(t)|^2 + |y(t)|^2 + 2|x(t)||y(t)| dt$$

$$\frac{1}{T}\int_{\langle T\rangle} |x(t)+y(t)|^2\,dt = \frac{1}{T}\int_{\langle T\rangle} \underbrace{|x(t)|^2}_{<\infty} + \underbrace{|y(t)|^2}_{<\infty} + \underbrace{2\,x(t)y(t)}_{<\infty}\,dt$$

$$< \infty$$

$\therefore\ x+y \in S_T$

ii) $(\alpha x)(t+T) = \alpha x(t+T) = \alpha x(t) = (\alpha x)(t)$

$$\frac{1}{T}\int_{\langle T\rangle} |(\alpha x)(t)|^2\,dt = \frac{|\alpha|^2}{T}\int_{\langle T\rangle} |x(t)|^2\,dt < \infty$$

$\Big\}\ \alpha x \in S_T$

$\therefore$ <u>S_T is a V.S.</u>